ASSIGNMENT - 9

AWS Educate and Amazon EC2: An IaaS Experience for Data Analysis

Practical Experience with Cloud Computing and Amazon EC2

1. Introduction

Cloud computing provides on-demand access to computing, storage, networking, and other information-technology resources without requiring the user to own and maintain the underlying physical infrastructure. Infrastructure as a Service (IaaS) is the cloud model in which users can provision and configure virtualized infrastructure such as compute instances, storage, and networking. Amazon Elastic Compute Cloud (Amazon EC2) is AWS's virtual-server service and is a central example of IaaS. AWS describes EC2 instances as virtual servers that can be configured with an operating system, applications, networking, storage, and security controls [1–3].

2. Objective

The objective of this activity was to obtain practical experience with cloud computing through AWS Educate and to understand how an IaaS EC2 environment can support computational data-analysis workloads. The practical work focused on launching and managing an EC2 instance, monitoring its state, configuring network security, remotely accessing the instance, resizing its compute resources, and testing termination protection.

3. Cloud Platform and Learning Environment

I used AWS Educate, specifically the “Getting Started with Compute” learning path. AWS Educate currently describes this course as providing hands-on experience with Amazon EC2 and cloud computing. AWS Educate provides guided learning activities and lab environments intended to let learners practice cloud skills without requiring them to build and maintain physical infrastructure [4].

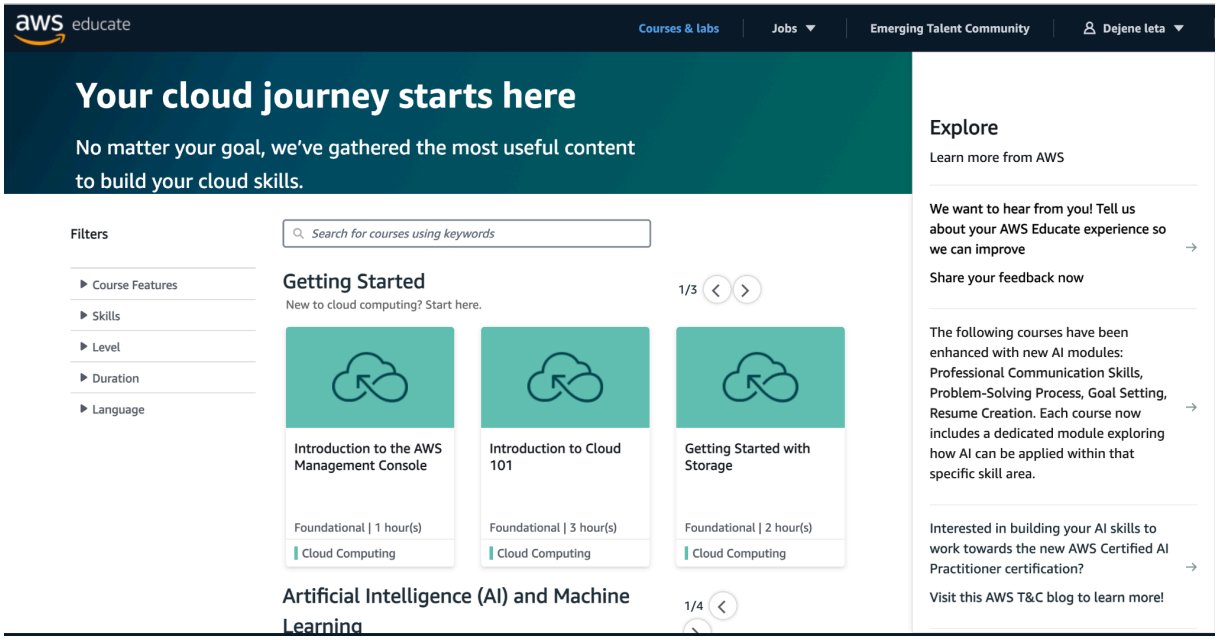


Figure 1. AWS Educate Courses & Labs page showing the available cloud-learning courses.

Figure 1 documents the AWS Educate environment used to access the cloud-computing learning material. The course was accessed through the AWS Educate interface rather than through a separately created personal AWS infrastructure account.

4. Selecting the EC2 / Compute Course

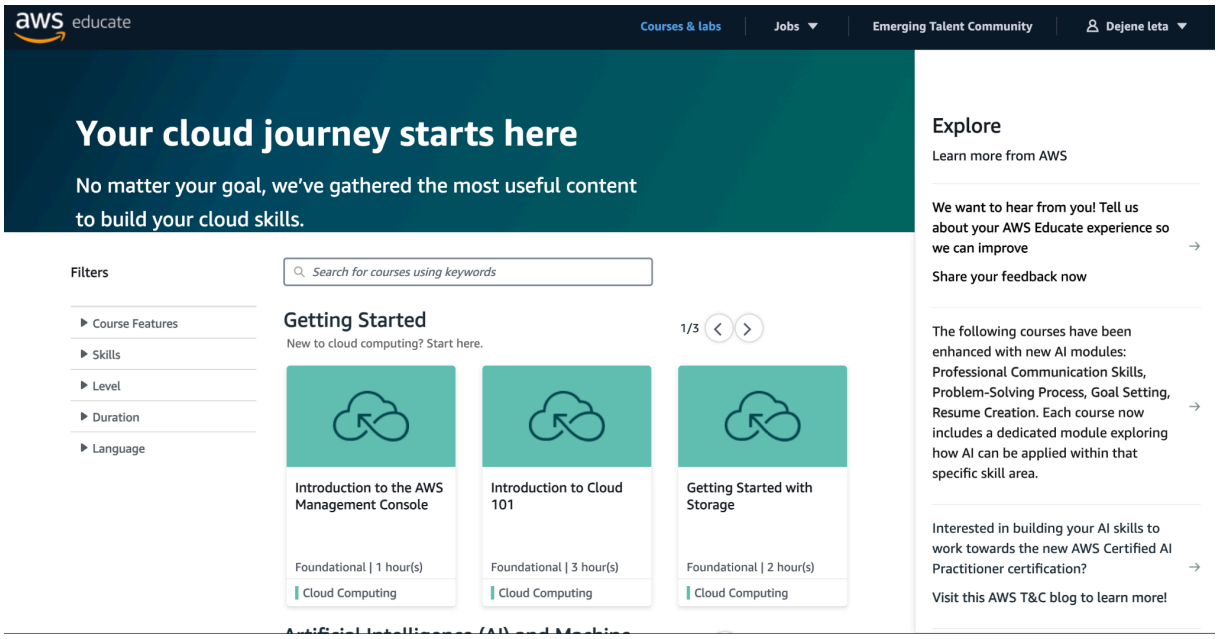


Figure 2. AWS Educate search results showing the "Getting Started with Compute" course and related EC2 resources.

The AWS Educate search interface was used to locate the compute course. “Getting Started with Compute” was selected because it specifically introduces Amazon EC2 and the management of cloud computing resources.

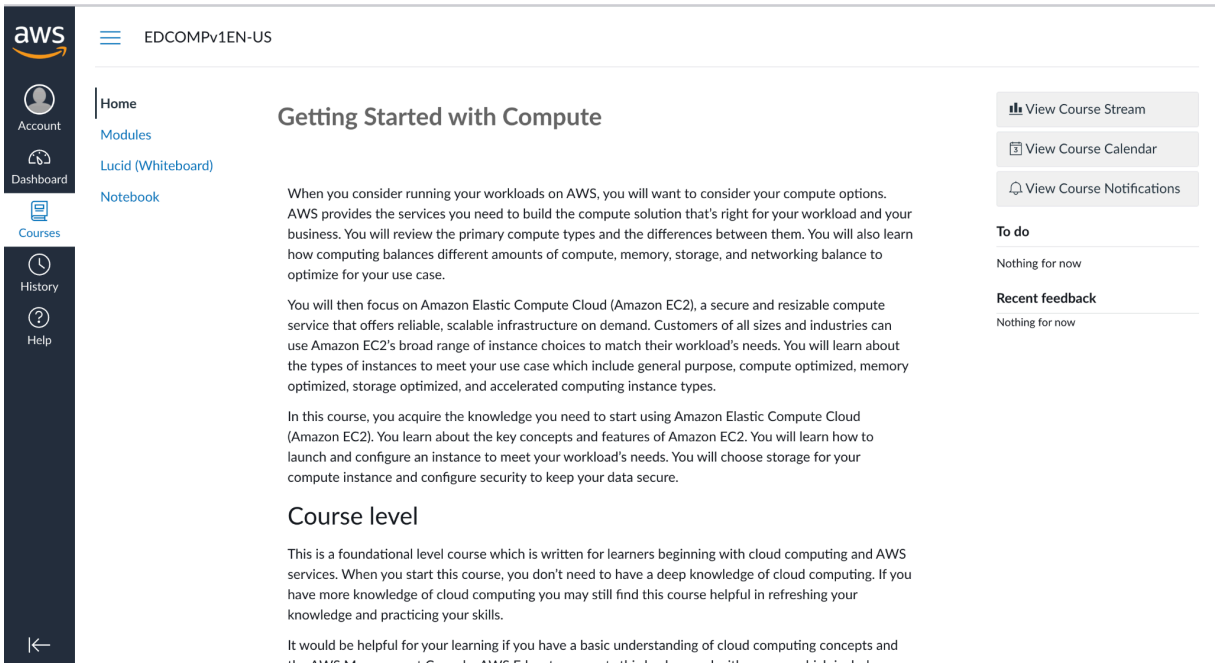


Figure 3. AWS Educate “Getting Started with Compute” course page introducing Amazon EC2 and cloud computing.

The course content explains the basic concepts required to work with AWS compute resources, including EC2 instances, instance choices, configuration, storage, and security.

5. Course Modules and Practical Simulation

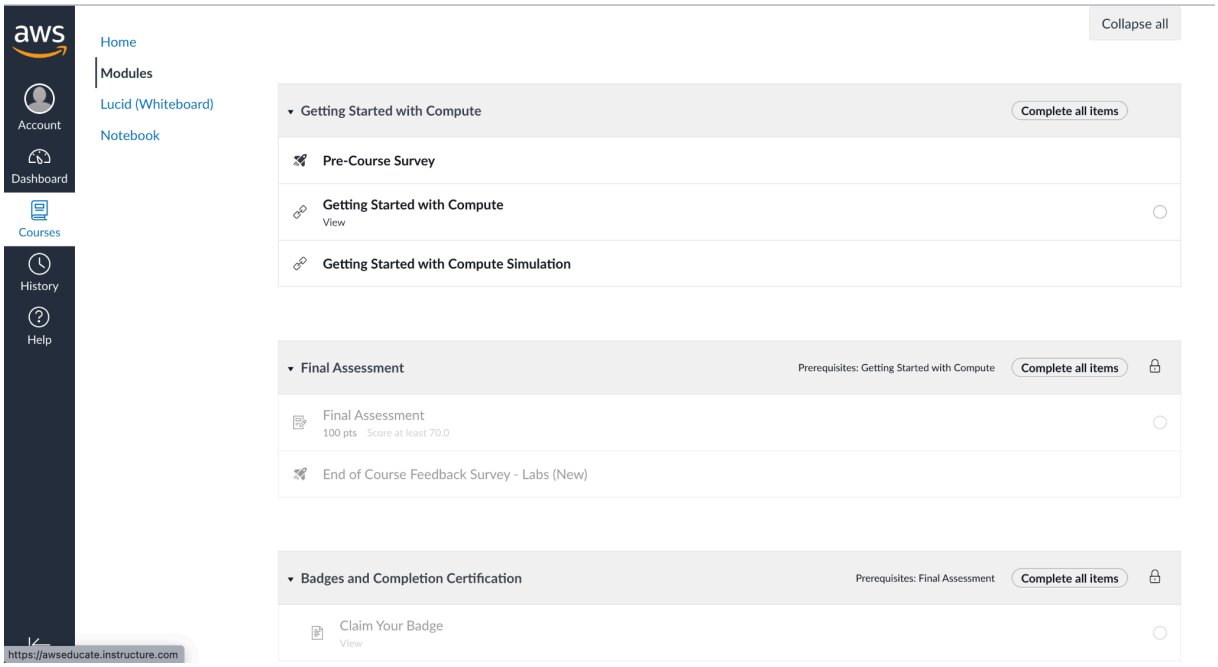


Figure 4. Course Modules page showing the Getting Started with Compute learning activities and final assessment.

The course was organized into introductory material, the compute lesson, an EC2 simulation, and a final assessment. The practical simulation was particularly important because it allowed the major EC2 lifecycle and management concepts to be practiced in a guided environment.

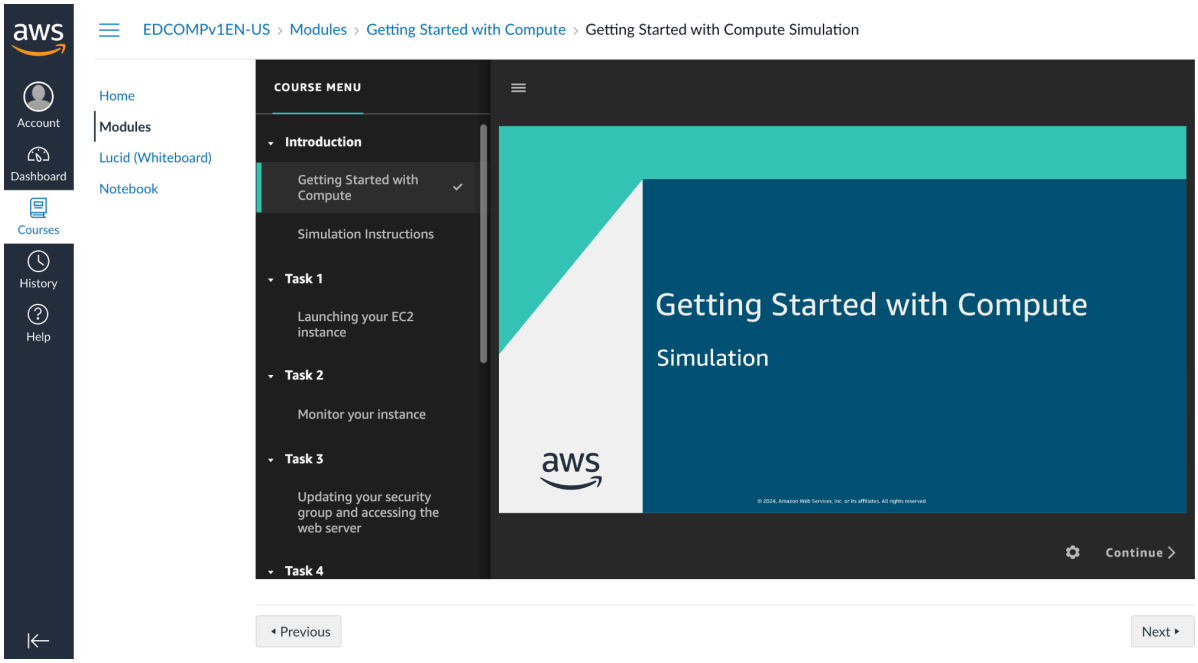


Figure 5. AWS Educate "Getting Started with Compute" simulation introduction and EC2 task sequence.

The simulation presented a sequence of EC2 management tasks. These tasks formed the main practical component of my cloud-computing experience.

6. EC2 Instance Launch and Configuration

The first practical activity involved launching an EC2 instance within the simulation. The instance was named “Web-Server.” The exercise demonstrated the role of an Amazon Machine Image (AMI), instance type, virtual network, security group, block storage, IAM-related configuration, and termination protection. AWS documentation describes an EC2 instance as a virtual server launched from an AMI, with the AMI providing the operating-system and application template [1,2].

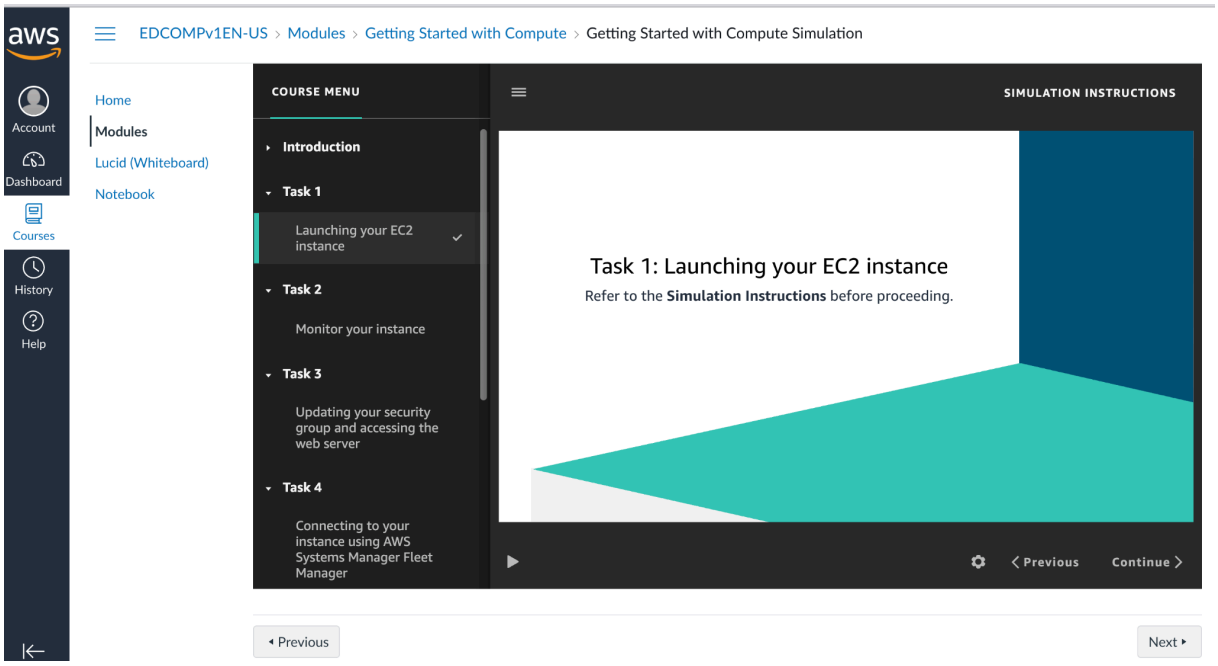


Figure 6. EC2 simulation task for launching the Web-Server instance.

The launch exercise showed that cloud infrastructure can be provisioned through a management interface rather than by purchasing, installing, and physically configuring a server.

7. Monitoring the EC2 Instance

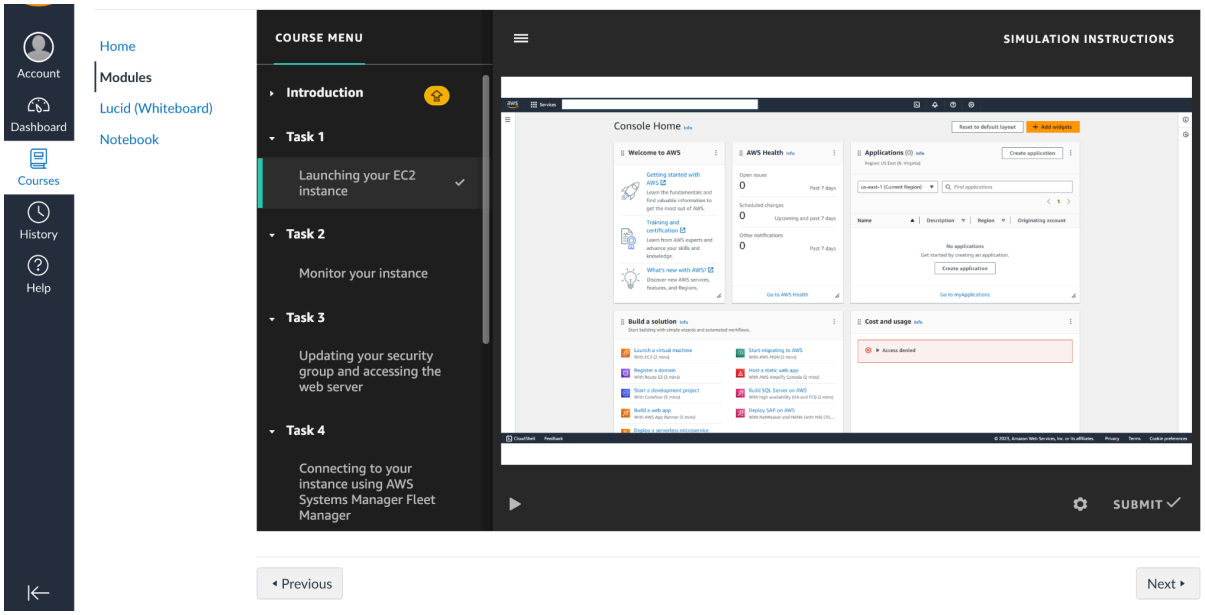


Figure 7. EC2 simulation showing the instance configuration during the launch workflow.

The monitoring task demonstrated how an EC2 instance can be inspected after launch. The simulation displayed instance state and configuration information, illustrating the importance of checking whether a computational resource is running correctly before using it for a workload.

8. Network Security and Web Access

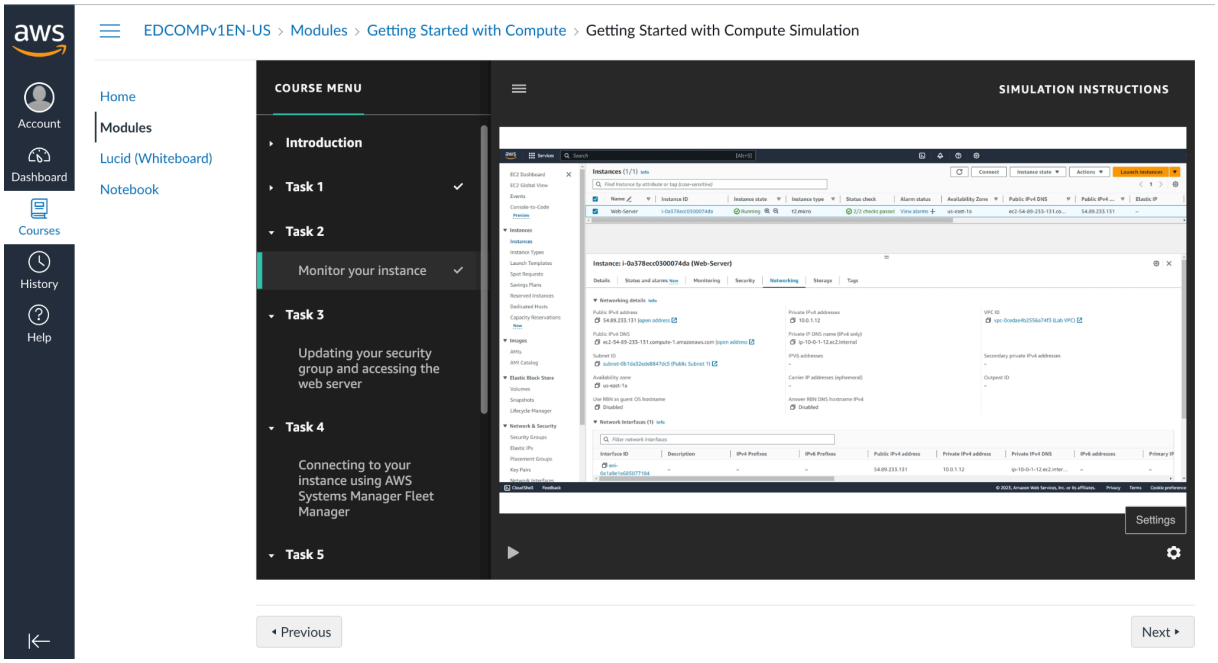


Figure 8. EC2 simulation showing the Web-Server instance and its configuration/monitoring information.

The security-group portion of the exercise demonstrated the role of a security group as a virtual firewall. The simulation used an inbound HTTP rule on TCP port 80 for the web-server exercise. This illustrated how network access can be controlled at the instance level rather than exposing every service to unrestricted traffic.

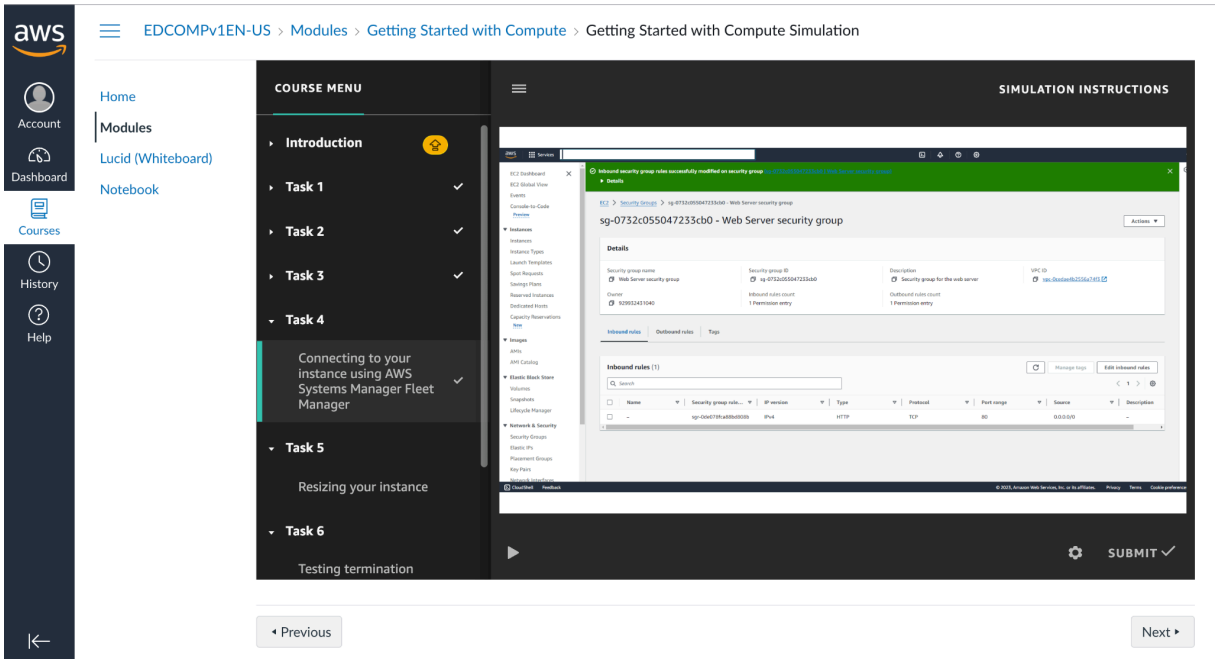


Figure 9. EC2 simulation showing the web-server/security configuration and progression through the practical tasks.

9. Remote Management

The simulation also introduced remote management through AWS Systems Manager Fleet Manager. This demonstrated that cloud-hosted computing resources can be administered remotely, which is particularly useful when computational workloads are running on servers that are not physically accessible to the analyst.

10. Resizing and Resource Scalability

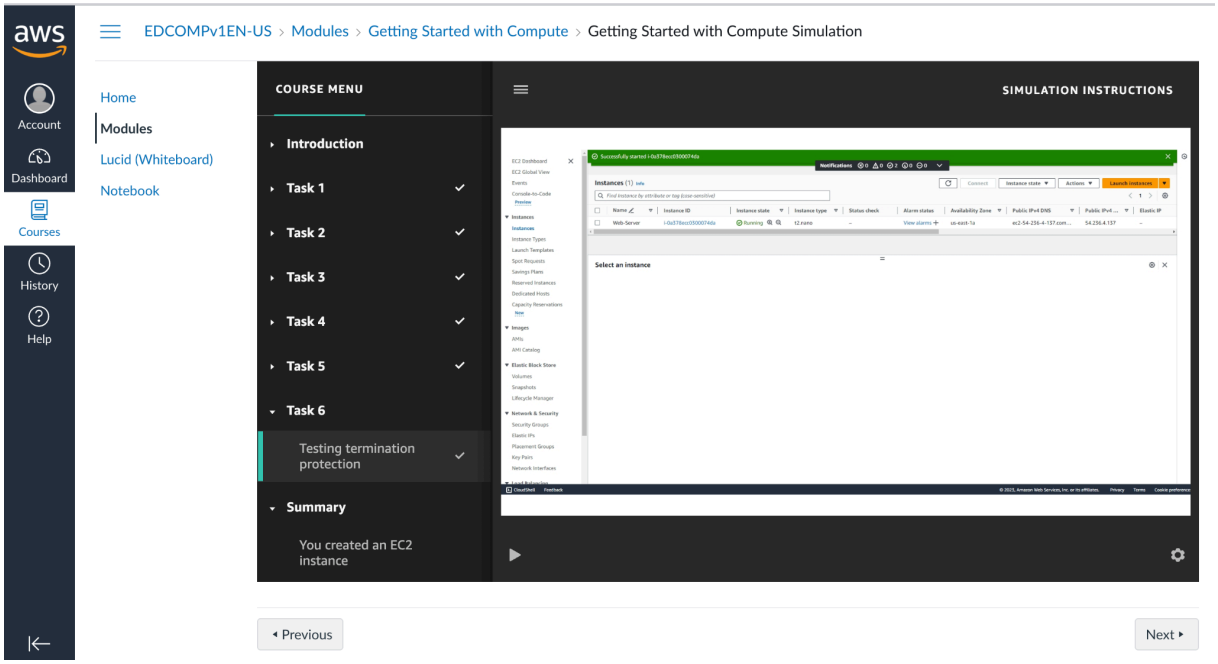


Figure 10. EC2 simulation showing successful progress through the instance-management tasks.

The resizing task demonstrated an important IaaS capability: computational resources can be changed according to workload requirements. For bioinformatics, this principle is useful because different analyses have very different CPU, memory, storage, and runtime requirements. For example, a small sequence-statistics script requires little compute capacity, whereas genome assembly, metagenomic analysis, or large-scale variant calling can require substantially more resources.

11. Instance Lifecycle and Termination Protection

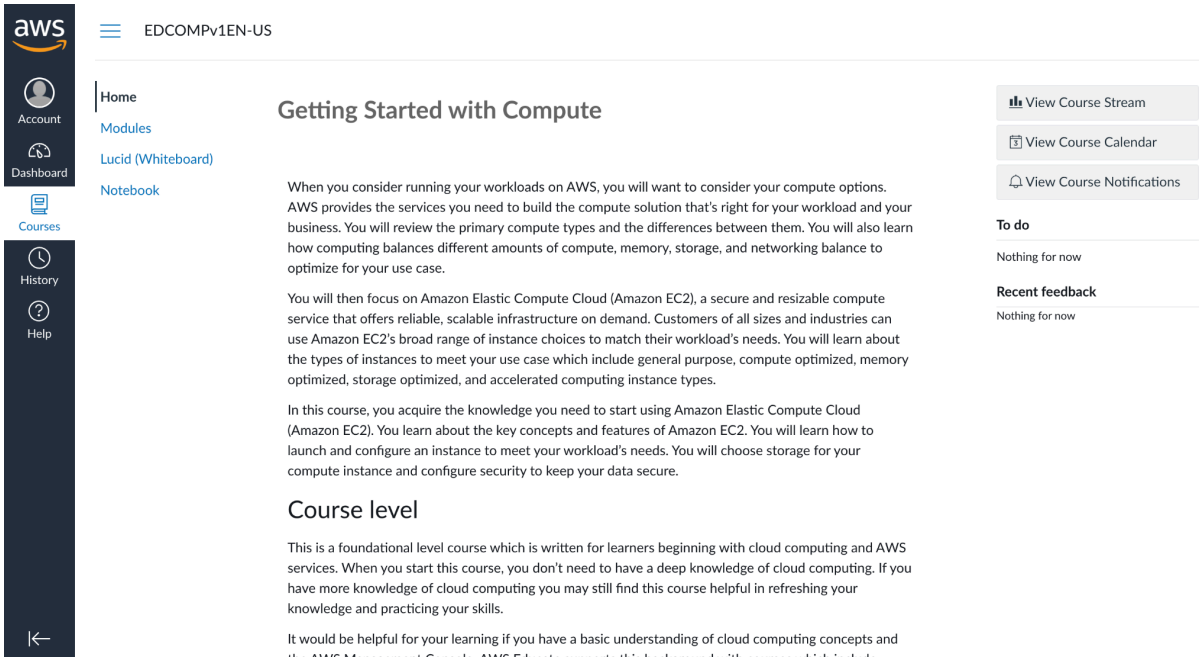


Figure 11. EC2 simulation summary showing completion of the practical task sequence.

The final practical task tested termination protection and completed the EC2 management sequence. This was important because instance lifecycle management is part of responsible cloud use. Users need to understand when resources are running, when they should be stopped or terminated, and how protective settings can reduce the risk of accidental deletion.

12. Course Completion and Assessment

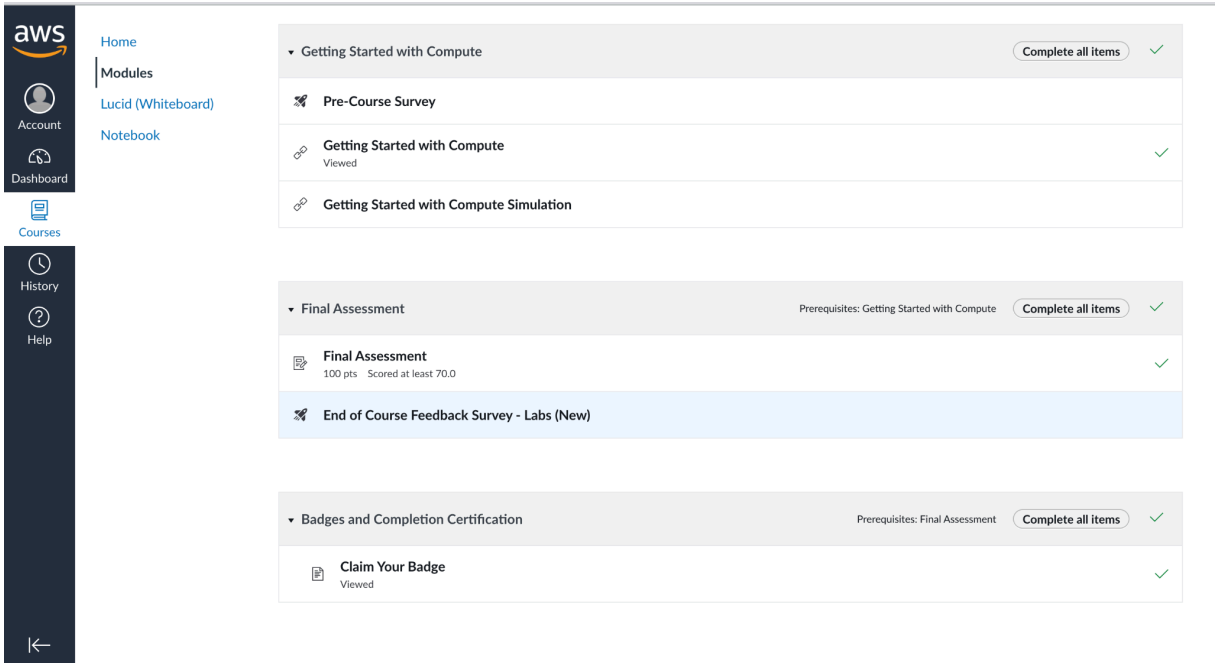


Figure 12. AWS Educate Modules page showing completion of the final assessment and the course completion/certification section.

The final course screen shows completion of the final assessment and progression through the completion/certification section. This provides evidence that the EC2 learning activity was not limited to reading introductory material; the course included a practical simulation followed by assessment.

13. EC2 as IaaS for Bioinformatics and Data Analysis

The main value of EC2 for bioinformatics is the ability to obtain a configurable computing environment on demand. In an IaaS model, the researcher does not need to purchase a physical workstation or server before beginning an analysis. Instead, the researcher can select an appropriate virtual machine, configure the operating system and software, attach storage, control network access, execute the analysis, and scale resources when necessary. AWS identifies compute services as an IaaS use case and describes EC2 as providing scalable virtual computing capacity [2,3].

14. Application to a Bioinformatics Workflow

A representative bioinformatics workflow that could be deployed on an EC2 Linux instance would be:

- Upload or retrieve biological data such as FASTA/FASTQ files.
- Create a Linux-based computational environment and install Python, Biopython, FastQC, or other required tools.
- Perform sequence quality control and preprocessing.
- Run sequence analysis, alignment, assembly, annotation, or statistical analysis depending on the research question.
- Store intermediate and final results on cloud storage or attached block storage.
- Scale the EC2 instance when the workload requires additional CPU or memory.
- Stop or terminate resources when the computation is complete, while retaining required persistent data.

These steps are presented as a proposed application of the EC2 infrastructure demonstrated in the course; they are not claimed as analyses that were executed during the AWS Educate simulation.

15. Experience and Reflection

My experience with AWS Educate changed my understanding of cloud computing from a mainly theoretical concept to a practical infrastructure-management process. Before the activity, an EC2 instance could be understood simply as a remote computer. The simulation demonstrated that an EC2 workload involves several interconnected components: compute capacity, an operating-system image, storage, networking, access control, remote management, monitoring, and lifecycle management.

The most relevant lesson for bioinformatics was scalability. Computational biology workloads are heterogeneous: some analyses can run on modest hardware, while others require substantial memory, CPU capacity, storage, or parallel execution. The ability to change instance resources provides a flexible alternative to maintaining a permanently oversized physical workstation.

Security was another important lesson. The security-group exercise demonstrated that computational resources should not simply be exposed to the network. Access should be limited to the ports and services required for the workload, and remote administration should be performed using appropriate management mechanisms.

16. Limitations of the Practical Activity

An important limitation is that the EC2 work completed for this assignment was performed through the AWS Educate guided simulation environment. The simulation reproduced the EC2 management workflow, but it did not constitute an unrestricted personal AWS EC2 server on which I independently executed a

biological dataset-analysis program. Therefore, this report does not fabricate a biological dataset, software output, runtime measurement, or computational result.

This distinction is important for academic integrity. The screenshots document what was actually completed: EC2 provisioning concepts, monitoring, security configuration, remote management, resizing, lifecycle management, and course assessment. The bioinformatics workflow described above is a realistic application of these IaaS capabilities, but it is clearly identified as a proposed application rather than a completed experiment.

17. Conclusion

The AWS Educate Getting Started with Compute course provided practical exposure to Amazon EC2 and the principles of Infrastructure as a Service. The completed activities demonstrated how a virtual computing instance can be launched, monitored, secured, remotely managed, resized, and protected against accidental termination. These capabilities are directly relevant to bioinformatics because computational analyses often require flexible combinations of CPU, memory, storage, networking, and software environments.

Although the completed simulation did not involve running an independent biological dataset analysis, it provided the infrastructure-level knowledge needed to understand how such analyses can be deployed in the cloud. Overall, the activity demonstrated the potential of EC2 to provide flexible and scalable computational infrastructure for future bioinformatics and data-science workloads.

18. References

1. Amazon Web Services. (2026). AWS Educate: Cloud skills for education. https://aws.amazon.com/education/awseducate/
2. Amazon Web Services. (2026). What is IaaS (Infrastructure as a Service)? https://aws.amazon.com/what-is/iaas/
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